

Predictive and Prescriptive Healthcare Analytics

Executive Summary

This project extends the descriptive and diagnostic healthcare model into predictive and prescriptive analysis. It uses 1,500 encounters recorded from January 2019 through December 2021. The integrated Power BI page combines a six-month revenue forecast, a multiple linear regression model, an AI Key Influencers visual, and two What-If parameters for cost reduction and length of stay (LOS). TotalCharges is treated as a proxy for Total Revenue because actual cash collections are not available.

The baseline contains \$36.29 million in billed revenue, \$21.92 million in cost, and a 39.61% margin. The regression model closely tracked the 2021 test period, with R2 of 0.977 and MAPE of 1.98%. Under the selected scenario, a 5% direct cost reduction combined with a 5% LOS reduction lowers estimated cost to \$19.78 million, releases about 572 bed-days, and raises the margin to 45.50%. These results support a controlled pilot rather than immediate system-wide implementation because the scenario assumes constant revenue and proportional cost changes.

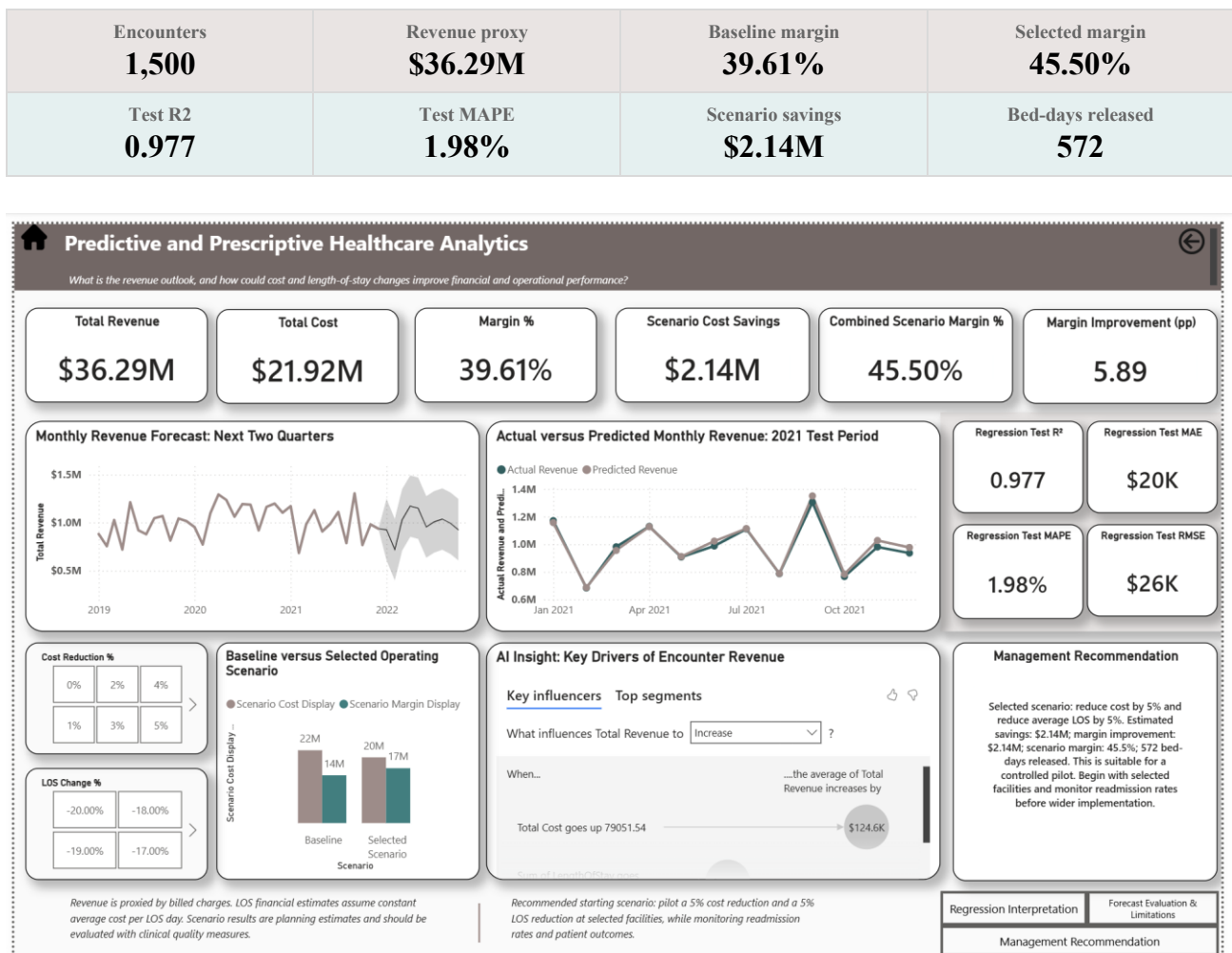


Figure 1 Integrated predictive and prescriptive healthcare dashboard

Data Preparation and Forecasting

Data and Model Design

The Power BI model retains the Assignment 2 star schema. FactEncounter contains one row per encounter and connects through one-to-many relationships to the date, patient, provider, facility, diagnosis, procedure, and payer dimensions. The source contains 500 patient records; 479 patients appear in the encounter fact table. AdmissionDateID supplies the monthly time key used for forecasting and regression.

Power Query preparation standardized data types, validated dimension keys, and retained the original encounter-level amounts. Measures calculate revenue, cost, margin, encounter count, average LOS, and readmission rate at query time. Separate parameter tables hold Cost Reduction % and LOS Change %, while a monthly analytical table supports regression training and testing. This structure preserves the fact table's encounter grain and avoids placing scenario values in the clinical dimensions.

Model component	Grain	Analytical role
FactEncounter	1,500 encounters	Charges, cost, LOS, readmission and foreign keys
DimDate	1,096 dates	Monthly ordering and historical period filters
Patient and provider dimensions	Patient or provider	Demographic and specialty context
Facility and clinical dimensions	Facility or category	Operational, diagnosis, procedure and payer context
What-If parameter tables	One selected value	Cost and LOS scenario assumptions

Time Series Forecasting Method

Monthly Total Revenue was plotted on a continuous date axis and Power BI's built-in line-chart forecast was configured for six monthly points, equivalent to the next two quarters. Power BI allows the forecast horizon and confidence interval to be set in the Analytics pane [1]. The central path remains broadly within the recent monthly range, approximately \$0.9 million to \$1.2 million, rather than showing sustained growth or decline. The shaded interval widens across the horizon, indicating that uncertainty increases further into 2022.

Forecast Evaluation and Limitations

The forecast is useful as a short-term planning range, but its accuracy has not been independently validated. The Power BI visual does not expose MAE, RMSE, or MAPE for the forecast, and no separate holdout result was reported. Performance was therefore evaluated qualitatively from the historical variation, the central forecast path, and the width of the confidence interval.

Only 36 monthly observations and three annual cycles are available, limiting stable estimation of seasonality.

The forecast is univariate and does not directly include payer mix, facility activity, future costs, inflation, or policy changes.

Billed charges may differ from actual collected revenue.

Sudden operational disruptions or structural changes may not follow the historical pattern.

The widening interval makes the output more suitable for short-term planning than precise budgeting.

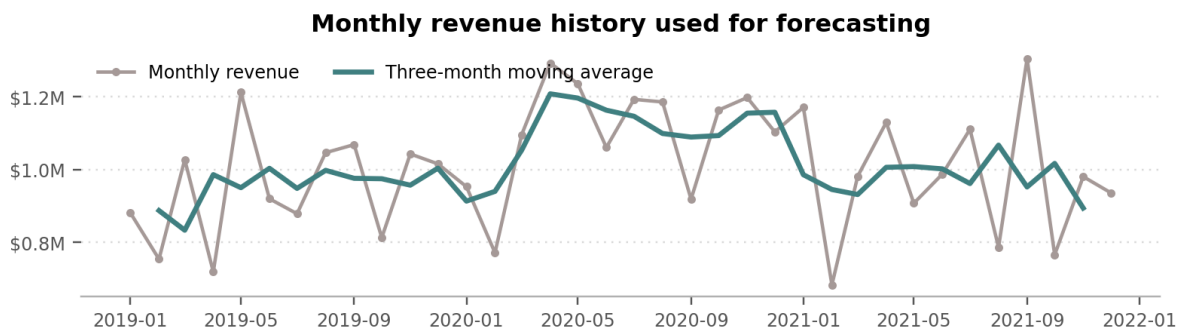


Figure 2 Historical monthly revenue supplied to the Power BI forecast

Regression Model and AI Analysis

Regression Method

A multiple linear regression predicts monthly billed revenue from monthly Total Cost and Encounter Count. The 24 months in 2019 and 2020 form the training sample, and the 12 months in 2021 form the test sample. The DAX LINESTX function estimates the least-squares line and returns coefficients and regression statistics [2]. The fitted equation is:

$$\text{Predicted Revenue} = -\$17,571.95 + 1.362934(\text{Total Cost}) + \$4,805.91(\text{Encounter Count})$$

Test R2	Test MAE	Test MAPE	Test RMSE
0.977	\$19,943	1.98%	\$26,412

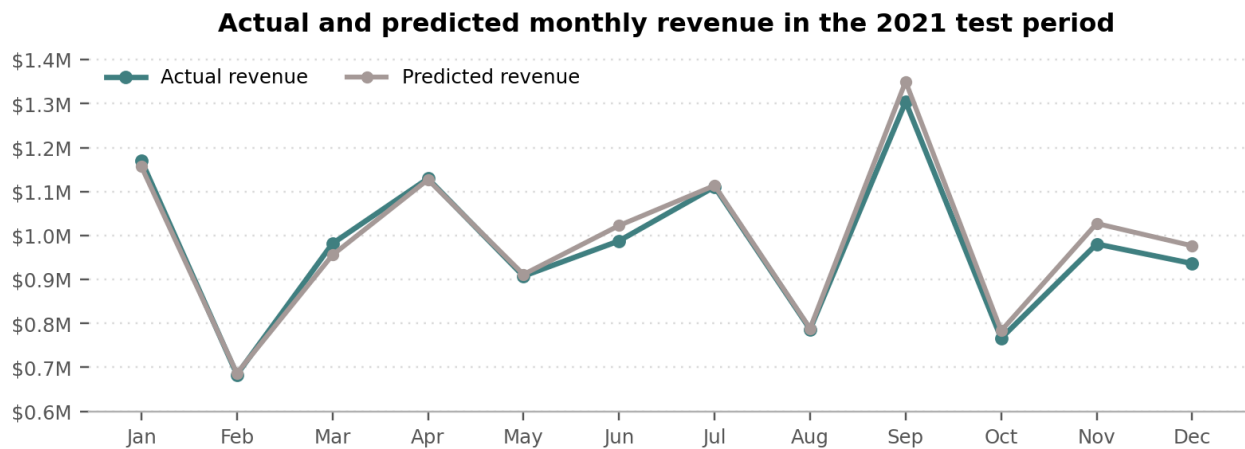


Figure 3 Regression predictions compared with actual monthly revenue

Performance Interpretation

The test R2 means that the model explains 97.7% of the observed variation in 2021 monthly revenue; it does not mean that predictions are 97.7% accurate. The MAE indicates a typical absolute monthly error of about \$20,000, and the MAPE places the average error at 1.98% of actual revenue. RMSE is higher than MAE because a few months, particularly September through December, have larger errors. The chart still shows close agreement across the full test period.

Holding encounter count constant, an additional dollar of monthly cost is associated with about \$1.36 in billed revenue. Holding monthly cost constant, one additional encounter is associated with approximately \$4,806 in revenue. These coefficients describe conditional associations and should not be interpreted as causal effects.

AI Key Influencers

The Key Influencers visual ranks the factors associated with higher encounter revenue [4]. In the displayed result, an increase of about \$79,052 in Total Cost is associated with an average revenue increase of approximately \$124,600. This supports the regression finding that cost and activity move with billed revenue, but it does not justify increasing costs to create revenue. The analysis contains only 36 monthly observations, and cost and revenue originate from the same encounter activity; both conditions can produce a very strong relationship.

Prescriptive What If Analysis

Scenario Design

Two numeric What-If parameters allow management to test direct cost reduction and percentage change in average LOS. Power BI parameters store the selected values and measures use those values to update report visuals [3]. Cost and LOS effects are applied sequentially to avoid double counting. Revenue is held constant, and LOS-related cost is assumed to change in proportion to LOS.

Measure	Logic
Scenario Cost	Baseline Cost x (1 - Cost Reduction) x (1 - LOS Reduction)
Scenario Margin	Total Revenue - Scenario Cost
Scenario Margin %	Scenario Margin / Total Revenue
Bed-Days Released	Total LOS Days x LOS Reduction

Scenario Results

Scenario	Cost red.	LOS red.	Cost (\$M)	Savings (\$M)	Margin %	Bed-days
Baseline	0%	0%	21.92	0.00	39.61%	0
5% cost only	5%	0%	20.82	1.10	42.63%	0
5% LOS only	0%	5%	20.82	1.10	42.63%	572
Selected 5% + 5%	5%	5%	19.78	2.14	45.50%	572
Higher 10% + 10%	10%	10%	17.75	4.16	51.09%	1,143

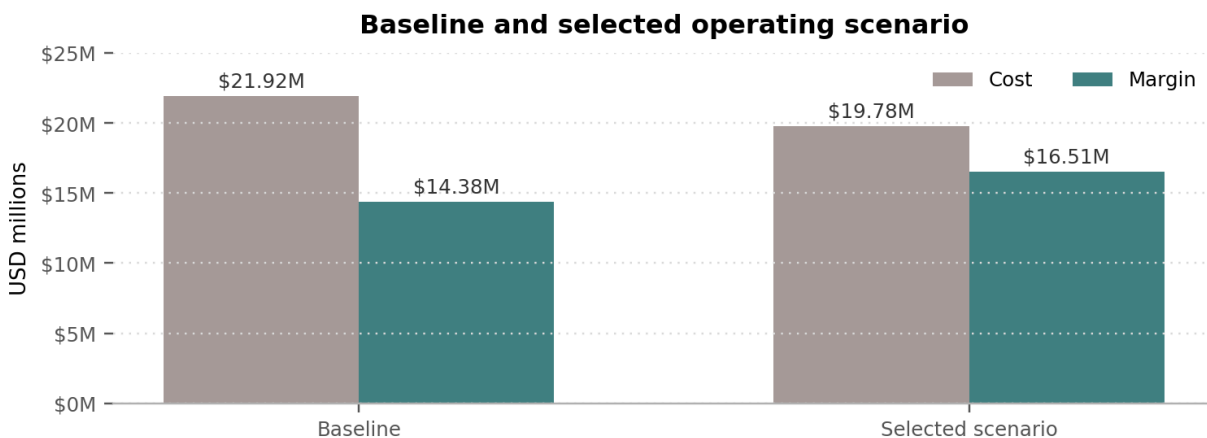


Figure 4 Baseline compared with the selected 5% cost and 5% LOS scenario

Selected Scenario Interpretation

At the selected 5% cost reduction and 5% LOS reduction, estimated total cost falls from \$21.92 million to \$19.78 million. The resulting \$2.14 million cost saving increases margin from \$14.38 million to \$16.51 million and raises margin percentage from 39.61% to 45.50%, an improvement of 5.89 percentage points. Average LOS falls from 7.62 to approximately 7.24 days, releasing about 572 bed-days. Cost savings and margin improvement are equal because scenario revenue is held constant.

The 10% plus 10% case illustrates the model's sensitivity but is not the recommended starting point. It assumes that fixed and variable costs respond proportionally and that shorter stays do not affect readmissions, complications, or revenue. Those assumptions require operational testing before a more aggressive target can be justified.

Management Recommendation and Conclusions

Recommended Course of Action

Hospital management should begin with the selected 5% cost reduction and 5% LOS reduction as a controlled pilot at a limited number of facilities. This scenario provides a meaningful estimated financial improvement without adopting the strongest assumptions in the sensitivity analysis. The pilot should target facilities with above-average cost or LOS and sufficient encounter volume for comparison. Facility-specific clinical baselines must be established before implementation.

Implementation Priorities

Select pilot facilities using baseline cost per encounter, average LOS, volume, and readmission performance.
Separate avoidable variable costs from fixed costs so that the savings target reflects costs management can influence.
Track actual cost, LOS, bed-days released, margin, readmissions, and patient outcomes during the same reporting period.
Compare observed results with the selected scenario and update the forecast and regression when additional monthly data become available.

Pilot Monitoring Measures

Measure	Purpose
Cost per encounter	Confirm that savings occur in costs management can influence
Average LOS and bed-days	Measure the operational effect of the LOS intervention
Margin percentage	Compare realized financial performance with the selected scenario
Readmissions and patient outcomes	Check that financial gains do not coincide with poorer care

Decision Guardrails

The overall historical readmission rate is 15.27%, but the pilot should use each facility's own baseline rather than treating the organization-wide rate as a universal threshold. Any apparent financial gain should be reviewed alongside readmission, complications, and patient experience. Management should pause or revise the intervention if shorter stays coincide with poorer clinical outcomes or if realized savings are materially below the model estimate.

Overall Limitations

The historical period contains only 36 months and does not establish long-run seasonality or structural stability.
Billed charges are used as a revenue proxy and do not measure contractual adjustments, denials, or cash collection.
The regression test set contains only 12 months, and Total Cost is closely related to the same encounter activity that generates revenue.
What-If results assume constant revenue and proportional cost changes; they do not separate fixed and variable cost behaviour.
The forecast, regression, and Key Influencers outputs identify patterns and associations, not causal effects.

Conclusion

The integrated dashboard satisfies the assignment's predictive and prescriptive objectives through two predictive methods, interactive scenario testing, AI-assisted exploration, and a clear management recommendation. The regression provides strong test-period fit, while the forecast offers a cautious near-term revenue range. The selected What-If scenario indicates material savings and capacity gains, but the assumptions require validation through a monitored pilot before broader adoption.

References

- [1] Microsoft Learn. Use the Analytics pane in Power BI. <https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-analytics-pane>
- [2] Microsoft Learn. LINSTX function in DAX. <https://learn.microsoft.com/en-us/dax/linestx-function-dax>
- [3] Microsoft Learn. Create and use parameters to visualize variables in Power BI Desktop. <https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-what-if>
- [4] Microsoft Learn. Create Key Influencers visualizations. <https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-influencers>